

FOR EEE DEPARTMENT
1st Year 1st semester 2025
Course: Phy 1202

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GROUP AND WEEK WISE SESSIONAL CLASS SCHEDULE

Group	1 ST WEEK	2 ND WEEK	3 RD WEEK	4 TH WEEK
1(a)	1a	2a	3a	4a
1(b)	1b	2b	3b	4b
2(a)	2a	3a	4a	5a
2(b)	2b	3b	4b	5b
3(a)	3a	4a	5a	1a
3(b)	3b	4b	5b	1b
4(a)	4a	5a	1a	2a
4(b)	4b	5b	1b	2b
5(a)	5a	1a	2a	3a
5(b)	5b	1b	2b	3b

Experiment: 01(a)/(b)

Name of the Experiment:

DETERMINATION OF THE SPECIFIC HEAT OF A LIQUID BY THE METHOD OF COOLING

Theory:

When a hot body is placed in a constant temperature surrounding, it loses heat by (i) conduction (ii) convection and (iii) radiation.

If however the body be such a way that the heat lost by conduction and convection may be neglected, then the quantity of heat emitted by the body in given time (or the rate of loss of heat) depends only on (a) the temperature of the body and that of its surrounding (b) the area and the nature of the surface exposed, and is completely independent of nature of the liquid.

If the difference of temperature between the body and its surrounding is not large, then according to Newton's law of cooling, the rate of emission of heat is proportional to the temperature difference. This principle may be utilized in determining the specific heat of a liquid by observing the time taken by the liquid in cooling from one temperature to another.

Suppose a liquid of mass M_1 and specific heat S_1 is enclosed within a calorimeter of mass m and specific heat s . The thermal capacity of the system is $(M_1S_1 + ms)$. If the temperature of the liquid falls from θ_1 to θ_2 in t_1 seconds, then the average rate of loss of heat is $(M_1S_1 + ms)(\theta_1 - \theta_2)/t_1$.

If now the first liquid be replaced by an equal volume of a second liquid (say water) of known specific heat under similar conditions and if the time taken by this second liquid to cool through the same range of temperature (θ_1 to θ_2) be t_2 seconds, then the average rate of loss of heat is $(M_2S_2 + ms)(\theta_1 - \theta_2)/t_2$, where M_2 and S_2 are the mass and specific heat of the second liquid. Since the conditions are similar, these two rates are equal.

$$(M_1S_1 + ms)(\theta_1 - \theta_2)/t_1 = (M_2S_2 + ms)(\theta_1 - \theta_2)/t_2$$

$$\text{or, } S_1 = \frac{M_2S_2t_1 + ms(t_1 - t_2)}{M_1t_2}$$

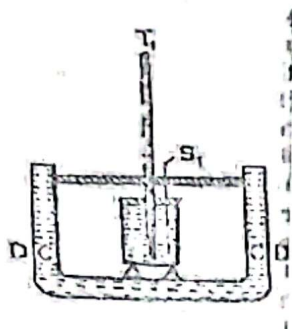


Fig1: Experimental setup for the determination of Newton's law of cooling.

Apparatus:

Double walled enclosure, Calorimeter, Thermometer graduated to a tenth of a degree, Heating bath, Stop watch.

Procedure:

1. Weight the cleaned and dried calorimeter with the stirrer.
2. Heat some water in a beaker to about 60°C (This temperature should be about 30°C above the room temp) and pour it in the calorimeter.
3. Gently stir the water and note its temperature at an interval of one minute.
4. Go on doing this till the temperature falls to about 35°C (which should be about 5°C above the room temp)
5. Now bring the calorimeter with its contents out side and wait till it cools to room temperature.
6. Remove the thermometer and weigh the calorimeter with water and stir and hence determine the mass of water.
7. Pour the water from the calorimeter and wipe it dry and put it back on its non-conducting support.
8. For liquid do the same operation described in (2-6).
9. Plot the two cooling curves (one for water and the other for liquid) on the same piece of graph paper with time as the abscissa and temperature as the ordinate (Fig.2).
10. From the curves determine the time taken by the water and the liquid to cool through the same range of temperature.
11. Calculate the specific heat from the formula given above.

Experimental Data:

Mass of calorimeter+stirrer, $m = \dots\text{gm}$

Specific heat of the material of the calorimeter, $s = \dots$

Mass of calorimeter + stirrer + water, $m' = \dots\dots\dots\text{gm}$

Mass of water, $M_2 = m' - m = \dots\dots\text{gm}$

Mass of calorimeter+stirrer+liquid, $m'' = \dots\dots\text{gm}$ Mass

of liquid, $M_1 = m'' - m = \dots\dots\text{gm}$

Specific heat of the liquid, $S_1 = \dots\dots$

Table1: Time-temperature record for water.

Time t_1 (min)									
Temperature($^{\circ}\text{C}$)									

Table 2: Time-temperature record for liquid.

Time t_2 (min)									
Temperature($^{\circ}\text{C}$)									

Calculations:

Data from graph,

Time taken by liquid to cool from $\theta_1^{\circ}\text{C}$ (.....) to $\theta_2^{\circ}\text{C}$ (.....) as obtained from the graph,
 t_1 = min.

Time taken by water to cool from $\theta_1^{\circ}\text{C}$ (.....) to $\theta_2^{\circ}\text{C}$ (... ..) as obtained from the graph,
 t_2 = min.

Specific heat of a given liquid,

$$S_1 = \frac{M_2 S_2 t_1 + m s (t_1 - t_2)}{M_1 t_2}$$

=

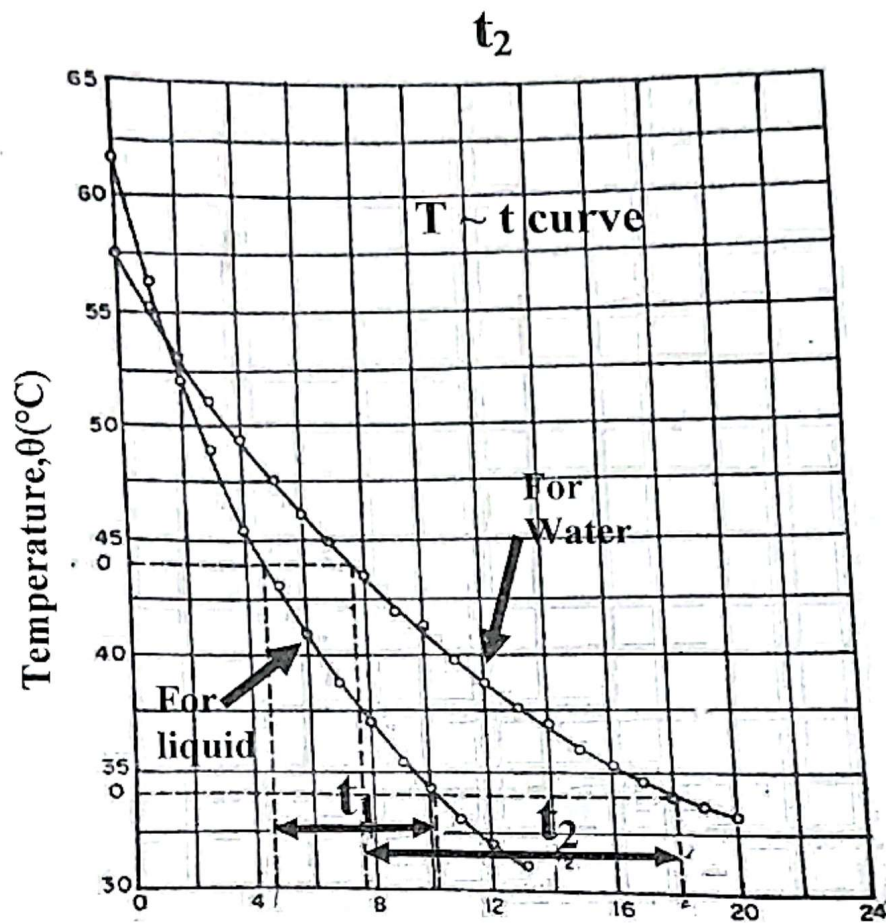


Fig2: Temperature Vs Time curve (cooling curves)

Results:

The specific heat of a liquid by the method of cooling is=

$$\text{Percentage of error} = \frac{\text{Expected value} - \text{Experimental value}}{\text{Expected value}} \times 100\% =$$

Precautions;

Discussions:

Experiment: 02(a)/(b)

Name of the Experiment:

TO DETERMINE THE CO-EFFICIENT OF THERMAL CONDUCTIVITY OF A METAL USING SEARLE'S APPARATUS

Theory:

If two points of a material of length d and area of cross section A are maintained at temperatures θ_1 and θ_2 where θ_1 is greater than θ_2 , then the amount of heat flowing from the hotter to the colder point in t seconds is given by

$$Q = \frac{KA(\theta_1 - \theta_2)t}{d} \dots\dots\dots(1)$$

where, K is the thermal conductivity of the material.

To measure the amount of Q , cold water is made to circulate round the colder ends through a coil of tube in contact with this end so that the quantity of heat Q raises the temperature from θ_3 to θ_4 of m gm of water flowing in t seconds. Then

$$Q = mS(\theta_4 - \theta_3) = m(\theta_4 - \theta_3) \dots\dots\dots(2)$$

If specific heat of water taken to be unity, then equation (1) and (2) it follows that

$$\frac{KA(\theta_1 - \theta_2)t}{d} = m(\theta_4 - \theta_3)$$

$$K = \frac{md(\theta_4 - \theta_3)}{A(\theta_1 - \theta_2)t} \dots\dots\dots \text{cal/cm/ sec/}^\circ\text{C.}$$

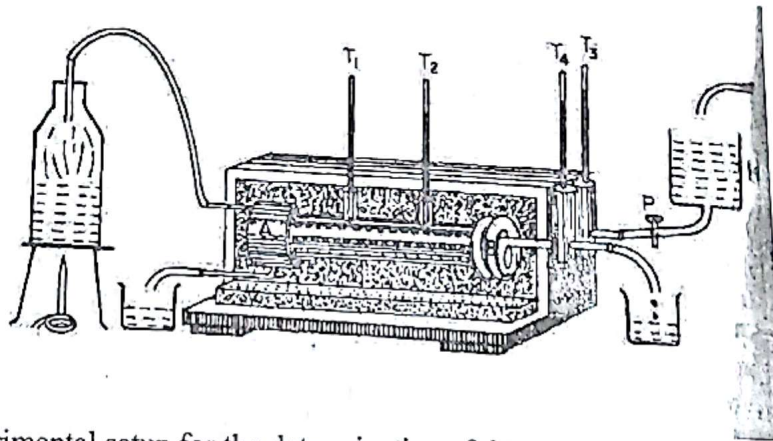


Fig 1: Experimental setup for the determination of the coefficient of thermal conductivity by Searle's apparatus.

Apparatus:

1. Searle's thermal conductivity apparatus,
2. Steam generators,
3. Four thermometers,
4. A beaker, stop-watch,
5. A balance,
6. A weight box and
7. Slide calipers.

Procedure:

1. Measure the diameter of the cylinder with slide calipers. Take a dry beaker and weigh it.
2. Heat the water boiler to get a continuous supply of steam to heat the end of the solid cylinder.
3. Connect the steam chamber to the boiler with rubber tubing. Connect also the outlet tube of the steam chamber with rubber tubing to a vessel containing water kept at some distance so that steam may escape into that vessel.
4. Connect the constant pressure head apparatus H. Adjust the flow of water from the tap by means of a pinch cock p.
5. Fit thermometers T_1 and T_2 in the holes of the cylinder and T_3 and T_4 in the water inlet and outlet tubes respectively.
6. Observe the readings in all the four thermometers. When the steady state is about to be attained, record the temperatures of all the four thermometers every two minutes intervals.
7. At the steady state (Temperature difference $T_3 \sim T_4$ is about 8°C - 10°C) collect water into the dry beaker for three minutes. The temperatures of the thermometers should remain constant during this collection of water. Weigh the beaker with the collected water.

Experimental Data:

Diameter of the cylinder =cm

Radius of the cylinder, r =cm

Area of cross section of the cylinder, $A = \pi r^2 = \dots\dots\dots$ sq. cm

Distance between the holes, d = cm

Mass of collected water, m =gm

Time for water collection, t =sec

Table 1: Data for thermometer readings (at steady state)

No. of obs.	Time (min)	$\theta_1(^{\circ}\text{C})$	$\theta_2(^{\circ}\text{C})$	$\theta_3(^{\circ}\text{C})$ Cold	$\theta_4(^{\circ}\text{C})$ Hot
1	0				
2	5				
3	10				
4	15				
5	20				
6	25				
7	30				
8	35				
9	40				
10	45				
11	50				
-	-				

Calculations:

The co-efficient of thermal conductivity, $K = \frac{md(\theta_4 - \theta_3)}{A(\theta_1 - \theta_2)t}$ cal/cm/ sec/ $^{\circ}\text{C}$.

Expected value =

$$\text{Percentage of error} = \frac{\text{Expected value} - \text{Experimental value}}{\text{Expected value}} \times 100\% =$$

Results: The co-efficient of thermal conductivity of the experimental metal =

Percentage of error =

Precautions:

Discussion:

N.B.: Precautions and Discussion may be three or more. Answer all the following questions.

Oral Questions:

1. Which types of conductors' thermal conductivity can measure using Searle's apparatus?
2. What are the factors that determine the thermal conductivity of a metal?
3. Which metal has the highest thermal conductivity?
4. Which method is used for measuring the coefficient of thermal conductivity of bad conductors?
5. Explain heat transfer by conduction procedure.

Experiment: 03(a)/(b)

Name of the Experiment:

DETERMINATION OF THE VALUE OF ACCELERATION DUE TO GRAVITY 'g' BY MEANS OF A COMPOUND PENDULUM

Theory:

A rigid body of any shape which can swing in a vertical plane about a horizontal axis passing through it is called a compound or physical pendulum.

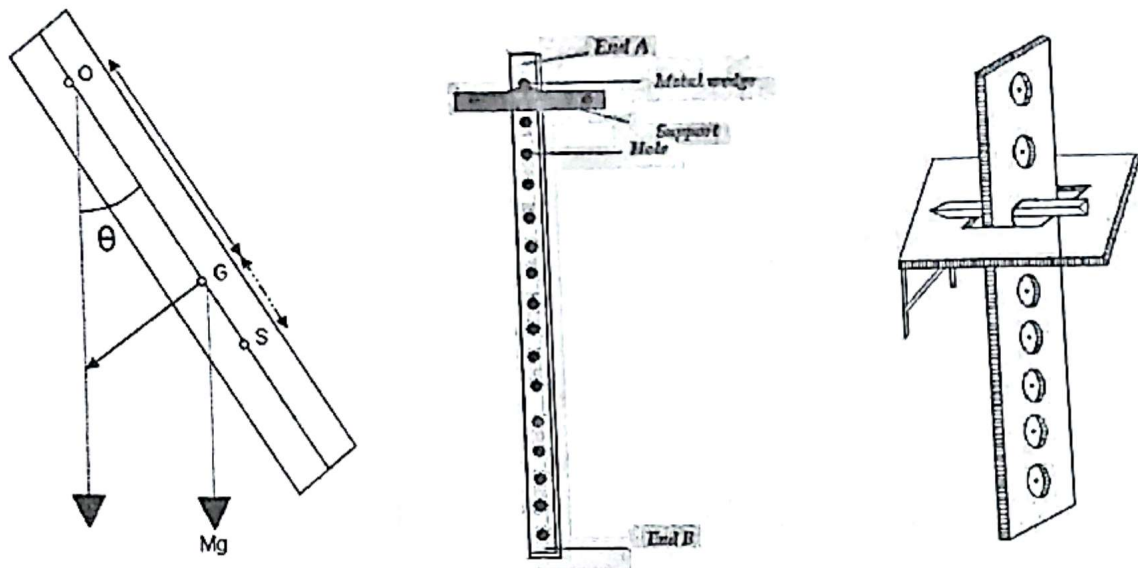


Fig.1: Compound pendulum

In Fig.1a compound pendulum is pivoted about a horizontal frictionless axis through P and is displaced from its equilibrium position by an angle, θ . In the equilibrium position the center of gravity G of the body is vertically below P. The distance GP is l and the mass of the body is m . The restoring torque for an angular displacement θ is,

$$\tau = -mgl \sin \theta \dots \dots \dots (1)$$

For small amplitudes, $\sin \theta = \theta$

$$I \frac{d^2\theta}{dt^2} = -mg \dots \dots \dots (2)$$

Where, I is the moment of inertia of the body through the axis P. Expression (2) represents a simple harmonic motion and hence the time period of oscillation is given by

$$T = 2\pi \sqrt{\frac{I}{mgl}} \dots \dots \dots (3)$$

If k is the radius of gyration of the pendulum about an axis through G parallel to the axis of oscillation through P , from the parallel axes theorem, $I = m(k^2 + l^2)$ and so

$$T = 2\pi \sqrt{\frac{k^2 + l^2}{gl}} \dots\dots\dots(4)$$

Equation (1) can be written in the quadratic form

$$l^2 - \frac{T^2 g}{4\pi^2} l + k^2 = 0 \dots\dots\dots(5)$$

If l_1 and l_2 are the two roots of equation (5) for a given time period T , then we can write

$$l_1 + l_2 = \frac{T^2 g}{4\pi^2} \dots\dots\dots(6)$$

$$l_1 l_2 = k^2 \dots\dots\dots(7)$$

As the sum and products of two roots are positive, the two roots are both positive. This means that there are two positions of the center of suspension on the same side of G about which the periods (T) would be same. Similarly, there will be two more points of suspension on the other side of the G , about which the time periods (T) will again be the same. Thus, there are altogether four points.

The distance between two such points (Suppose O and S in Fig. 1), asymmetrically situated on either side of the G will be the length (L) of the simple equivalent pendulum. The period of oscillation about either O or S is the same. The point O and S are called the center of suspension and center of oscillation respectively. These points are interchangeable. Centre of suspension is a point where the horizontal axes of rotation intersect the vertical section of the pendulum taken through center of gravity.

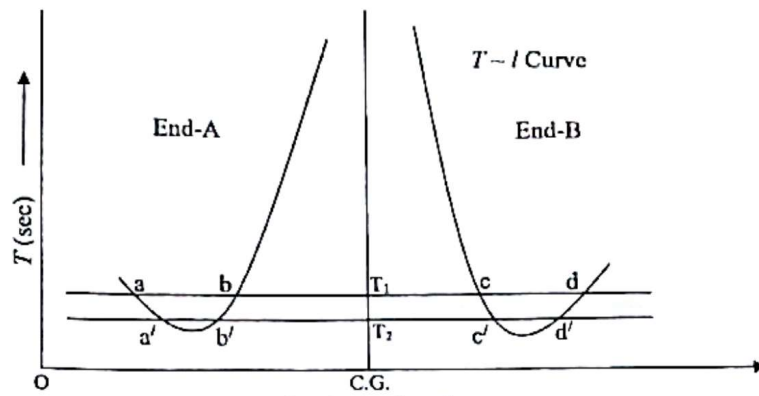
Equation 6 can be rearranged to get

$$g = \frac{4\pi^2(l_1 + l_2)}{T^2} \dots\dots\dots(8)$$

Obviously, $l_1 + l_2$ is the length of the equivalent simple pendulum. Thus, g can be calculated using Eq. 8. By finding L graphically, and determining the value of the period T , the acceleration due to gravity (g) at the place of the experiment can be determined.

Apparatus:

1. Compound pendulum,
2. Stop-watch,
3. Meter scale and
4. Balance with weight box



Time Period vs Distance from End A

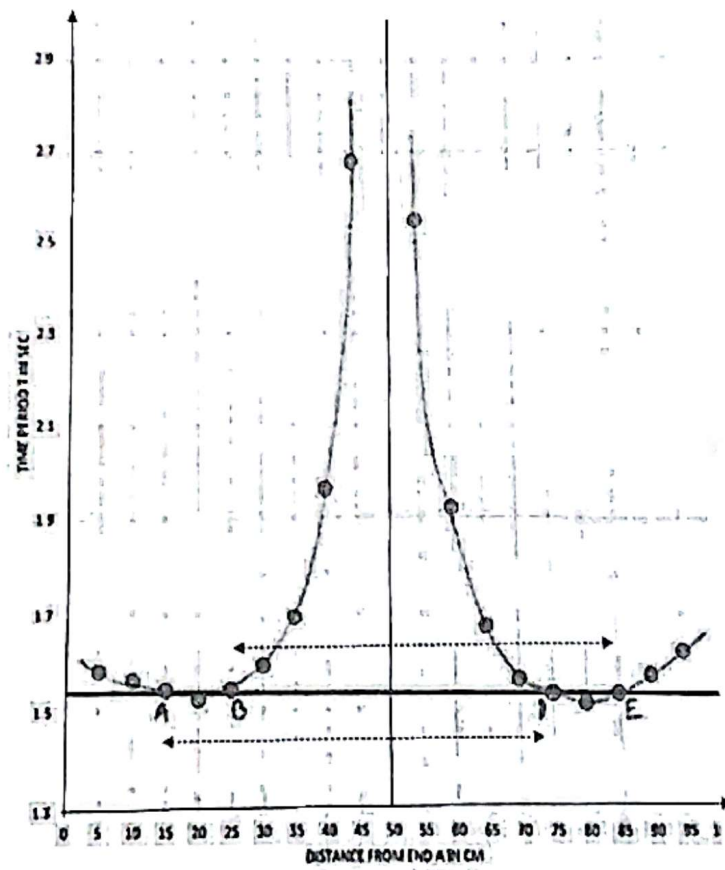


Fig.2: Time period versus distance from the center of gravity curve

Procedure:

1. Determine the center of gravity of the compound pendulum.
2. Determine the distance of each of the holes from the center of gravity.
3. Hang the pendulum and record the time for 15 oscillations (at least two times and take mean) for each of the holes (from End-A to End-B) as point of suspension.
4. Calculate the time period.
5. Plot a $T-l$ graph (Fig.2). Find the **equivalent length** corresponding to a certain time period T for both sides of the pendulum by drawing one line parallel to the l -axis. Calculate, 'g' for the line using the values of l_1+l_2 for line $abcd$ obtain from this process.

Experimental Data:

Table1: Data for time period, T

End A			End B	
Time period, T (sec)	Time for 15 oscillations, t (sec)	Hole number Or, Distance of knife- edge from fixed end	Time period, T (sec)	Time for 15 oscillations, t (sec)

Calculations:

Data from T-l curve

$$l_1+l_2=(ac+bd)/2= \quad \text{cm,}$$

Hence,

$$g = \frac{4\pi^2(l_1+l_2)}{T^2} = \quad \text{cm/Sec}^2$$

Expected Value=

Percentage of error $\frac{\text{Expected value} - \text{Experimental value}}{\text{Expected value}} \times 100\% =$

Results:

The value of 'g', Acceleration due to gravity by means of a compound pendulum as found in this experiment, $g = \text{ cm/sec}^2$
Percentage of error=

Precautions:

Discussion:

N.B.: Precautions and Discussion may be three or more. Answer all the following questions.

Oral Questions:

1. What is a compound pendulum? What are the defects of the compound pendulum?
2. Which is superior-- compound pendulum or a simple pendulum?
3. What do you mean by center of suspension and center of oscillation?
4. What is the length of the equivalent simple pendulum?
5. What is meant by G ? How are G and g related?
6. For how many points in a compound pendulum the time period is the same?
7. What are the drawbacks of a simple pendulum which have been overcome in a compound pendulum?

Experiment: 04(a)/(b)

Name of the Experiment:

DETERMINATION OF THE SURFACE TENSION OF WATER BY CAPILLARY TUBE METHOD

Theory:

The surface tension of a liquid is the force acting perpendicular to each centimeter of the imaginary line in the plane of the surface. If one end of a clean capillary tube of fine bore is dipped into a liquid, the liquid rises up the tube through a height h (Fig. 1), the surface tension T acts upwards along the tangent to the meniscus. The component of T acting vertically upwards $T \cos \theta$ and the total force acting upwards is $2\pi r \times T \cos \theta$, r being the internal radius of the capillary tube. This is the upward force due to surface tension of the liquid.

The weight of the liquid column acting downwards is equal to $V \times \rho \times g$

where ρ is the density of the liquid and g is the acceleration due to gravity.

The volume $V = \pi r^2 h + \text{volume of the meniscus}$ for a tube of uniform bore.

If the radius r is small, the meniscus = (volume of a cylinder of radius r and height h) – (volume of hemisphere of radius r).

This can be written as

$$\text{Volume of meniscus} = \pi r^3 - \frac{2}{3} \pi r^3 = \frac{1}{3} \pi r^3$$

$$\text{Therefore, } V = \pi r^2 h + \frac{1}{3} \pi r^2 h = \pi r^2 \left(h + \frac{r}{3} \right)$$

$$\text{Hence weight of the liquid column} = \pi r^2 \left(h + \frac{r}{3} \right) \rho g$$

Since the column is in equilibrium the upward force due to surface tension must support the weight of the liquid column.

$$\text{Hence, for equilibrium, } T \cos \theta \times 2\pi r = \pi r^2 \left(h + \frac{r}{3} \right) \rho g$$

For water is zero and hence $\cos \theta$ is unity. So the above relation gives

$$T = \frac{\rho g r}{2} \left(h + \frac{r}{3} \right) \text{ dynes/cm}$$

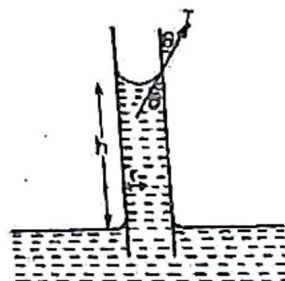


Fig 1: Surface tension by capillary tube.

Apparatus:

1. Capillary tubes of varying bores,
2. Travelling microscope,
3. Needle, Beaker,
4. Clamp and
5. Stand.

Procedure:

1. Take a capillary tube and clean the bore.
2. Make the base and the axis of a travelling microscope horizontal by using a spirit level. Also focus the cross wires of the microscope sharply.
3. Take some water in a clean beaker. Hold a glass plate G in a clamp and attach the capillary tube A to it with wax (Fig. 2). Put the glass plate G in such a position that the capillary tube is vertical and its ends dip well in the water. Water will rise in the tube.
4. Measure the length of the needle by meter scale. Place the needle by the sides of the tubes and adjust its position such that its lower point is just above the water surface in the beaker. If it dips in water it will rise along its sides.
5. Focus the microscope on the needle and adjust it such that the horizontal cross wire coincides with the top of the needle. Read the vertical scale of the microscope.
6. Adjust the height of the microscope such that its horizontal cross wire just touches the lower meniscus in tube A (see Fig. 2). It will look inverted; read the microscope scale. Take the difference of this reading and the reading in (5) above.
7. Make the vertical cross wire tangential to the left side of the bore. Read the horizontal scale of the microscope. Shift the microscope horizontally until the vertical cross wire is tangential to the right side of the bore. Similarly, by the vertical movement of the microscope, determine the diameter in the perpendicular position. Take the mean of these two diameters and then find out the radius.

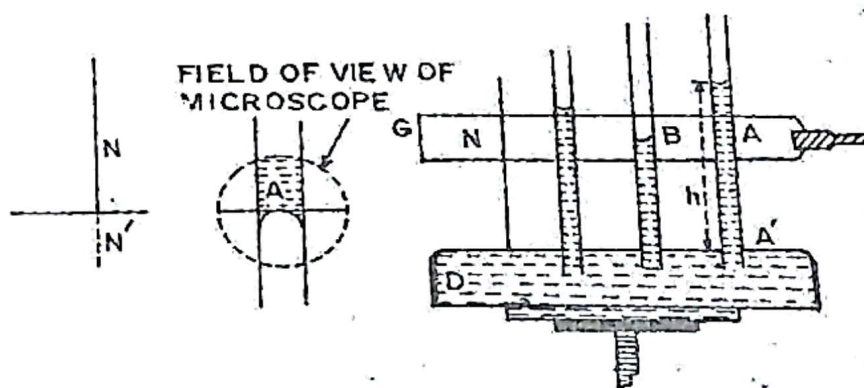


Fig 2: Experimental set up for the determination of surface tension by capillary tube.

Experimental data:

The length of the needle, $l = 8.5\text{cm}$

Density of water at 0°C , $\rho = \dots\dots\dots\text{gm/cm}^3$

Acceleration due to gravity, $g = \dots\dots\dots\text{cm/sec}^2$

Table 1: Data for the needle head (h_1)

No. of obs.	Readings for the needle head, (h_1) (cm)				
	Main Scale Reading (MSR) (a)	Vernier Scale Reading (VSD) (b)	Vernier Constant (VC)(c)	Total = $a+(b \times c)$	Mean (h_1)
1					
2					
3					

Table 2: Data for the water meniscus in the tube, (h_2)

No. of obs.	Readings for the water meniscus in the tube, (h_2) (cm)				
	Main Scale Reading (MSR) (a)	Vernier Scale Reading (VSD) (b)	Vernier Constant (VC)(c)	Total = $a+(b \times c)$	Mean (h_2)
1					
2					
3					

Table 3: Data for the radius for the capillary tube, r

Direction of observation	Readings for the bore (cm)							Difference ($r_1 \sim r_2$)	Mean diameter D (cm)	Radius r (cm)	
	Left or lower end				Right or upper end						
	MSR(a)	VSD(b)	VC (c)	Total (r_1) = $a+(b \times c)$	MSR(a)	VSD(b)	VC(c)				Total (r_2) = $a+(b \times c)$
Horizontal											
Vertical											

Calculations:

Difference of the readings at needle head and water meniscus in tube,

$$A = h_1 \sim h_2 = \dots\dots\dots \text{cm}$$

$$\text{Height of the water meniscus in tube, } h = (l \pm A) = \dots\dots\dots \text{cm}$$

Surface tension of water,

$$T = \frac{\rho g r}{2} \left(h + \frac{r}{3} \right) = \dots\dots \text{dynes/cm}$$

Expected value =

$$\text{Percentage of error} = \frac{\text{Expected value} - \text{Experimental value}}{\text{Expected value}} \times 100\% =$$

Results:

The surface tension, T of water by capillary tube method =

Percentage of error =

Precautions:

Discussion:

N.B.: Precautions and Discussion may be three or more. Answer all the following questions.

Oral Questions:

1. What is surface tension? What is its unit?
2. Define angle of contact
3. Explain cohesive and adhesive forces.
4. What is the effect of temperature on surface tension?
5. Why should water be free from grease?
6. On what factors does the surface tension of a liquid depend?
7. Will the liquid overflow if the tube be insufficient height?
8. Why should distilled water not be used?
9. Will all liquids rise in the capillary glass tubes?
10. How does the surface tension of pure water differ from that of salt solution?
11. We have seen that when a capillary glass tube is dipped into water the water level rises inside the tube. But when a wide-bore glass tube is dipped in water no rise is observed. Why?

EXPT NO: 5(a)(b)

Name of the Experiment:

VERIFICATION OF THE LAWS OF TRANSVERSE VIBRATION OF STRINGS AND DETERMINATION OF THE FREQUENCY OF A TUNING FORK BY MELDE'S EXPERIMENT

Theory:

Let one end B of the string, which may be a piece of stout thread or thin cord, be attached to one prong of the fork F. The fork is screwed vertically on a wooden base W at the edge of a table. The other end A passes over a small pulley and is attached to a scale pan S (Fig. 1).

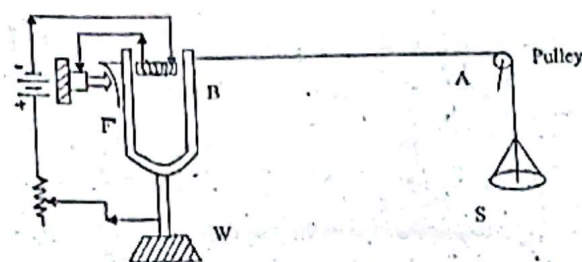


Fig. 1: Experimental arrangement of Melde's experiment

The string will be set into vibration by setting the tuning fork into vibration. As a result, waves will proceed along the length of the string and will be reflected back on the reaching the fixed end of the string. The superposition of the direct and reflected waves will form stationary waves in which the extreme fixed ends of the string will always be nodes and in between them there may be one or more antinodes (Fig. 1) depending on the tension or the length, the frequency n of the fork may be made to be equal to the frequency n' of the fundamental or any one of the higher tone of the string. When this happens, a resonance is said to have occurred between the fork and the particular mode of vibration of the string.

If the node of vibration be assumed to be fundamental then $\lambda = 2l$, where l is the length of the string. The frequency of the fork will then be given by the relation,

$$n = n' = \frac{1}{\lambda} \sqrt{\frac{t}{m}} = \frac{1}{2l} \sqrt{\frac{t}{m}}$$

Where, m is the mass per unit length of the vibrating string in gram and t is the tension applied to the string and is expressed in absolute unites, i.e., dynes or poundals. Now the motion of the prongs of the fork, which sets the string in resonance vibration, may be in two different direction- (i) in a direction perpendicular to the length of the string i.e., transverse position (Fig. 2a) and (ii) in a direction along (parallel) to the length of the string i.e., longitudinal position (Fig. 2b).

In the transverse case, the frequency n of the fork is the same as n' , the frequency of the string while in the longitudinal case it is twice as great. This is because in the longitudinal case, the vibration is produced by the alternation pulls upon the end of the string by the prong of the fork. Each movement of the prong to the right pulls the string tight, i.e., the string is stretched and this occurs in the middle of the swing, i.e.,

twice in every vibration. Thus, in this case, the frequency of the fork is half that of the fork or in other words, the frequency of the fork is twice the frequency of the string.

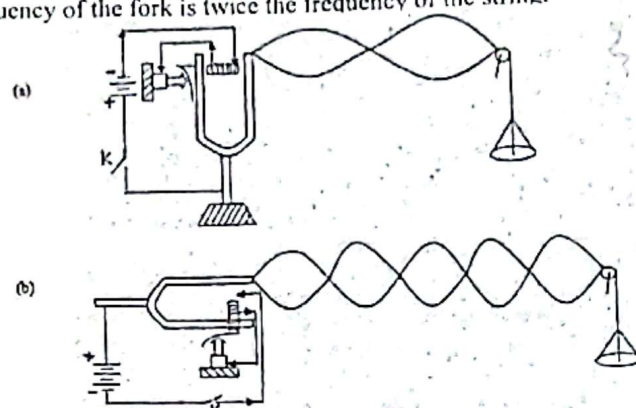


Fig. 2: Experimental arrangement for (a) longitudinal vibration (b) transverse vibration

Therefore, for transverse position, the frequency n of the fork is

$$n = n' = \frac{1}{\lambda} \sqrt{\frac{t}{m}} = \frac{1}{2l} \sqrt{\frac{t}{m}} \dots \dots \dots (1)$$

Where, l is the length of a segment or loop between two consecutive nodes of the string.

$$\text{Or, } \frac{t}{l^2} = 4mn^2 = \text{constant}$$

For longitudinal position, the frequency n of the fork is,

$$n = 2n' = \frac{2}{2l} \sqrt{\frac{t}{m}} = \frac{1}{l} \sqrt{\frac{t}{m}} \dots \dots \dots (2)$$

$$\text{Or, } \frac{t}{l^2} = 4mn^2 = \text{constant}$$

Thus, by altering the tension t and hence the wave length one should be able to determine the frequency of a tuning fork and also to find $t/l^2 = \text{constant}$ which verifies the laws of transverse vibrations in a string.

Apparatus:

1. Melde's apparatus,
2. Thread,
3. Wooden clamps,
4. Meter scale and
5. Weight box.

Procedure:

1. Weigh the scale pan.
2. Clamp the tuning fork in either transverse or longitudinal position at one edge of the table.
3. Fix a pulley over a clamp, screwed at other edge of the table.
4. Fasten a thread, about a meter long, to the top of the prong and pass the other end over the pulley.
5. Hang the scale pan to this end and put some weights (about 5 gm) on it so that the thread is lightly stretched.
6. Set the fork into vibration by a smart light blow with a rubber padded hammer. This will set the thread into vibration and nodes and loops, possibly ill defined.

7. Note the total number of loops along the entire length of the thread. Increase or decrease the weight, as the case may be, until the loops are well defined. The weight in which the loops are most well defined is the weight to be used in calculation for the given number of loops.
8. Determine the length between two successive nodes. To do this place two pins P_1 and P_2 of adjustable heights below the two extreme well defined nodes but not below the nodes at the two ends of the string.
9. Accurately measure the distance G between the pins and count the total number of loops N between them.
10. Repeat the operation thrice by independently adjusting the positions of the pointers.
11. Increase the weight in the scale pan by about 5 gm or so when the total number of loops between the two fixed ends become different. Note this number. Taking care that this number does not change.
12. Repeat the operations described in (6) to (10) to determine the new length l of a segment.
13. In this way go on increasing the weight on the scale pan in steps of about 5 gm. The number of loops will decrease with increasing weights.
14. Note the total number of loops for each new weight and determine the corresponding l in the manner described (6) to (10).
15. Repeat the whole process with the other mode of vibration of the fork. It may be mentioned that when the fork is in the transverse mode, more weight will have to be put on the pan to start with.
16. At the end of the experiment, accurately determine the weight of a known length of the thread with the help of a sensitive balance.
17. Hence determine the mass per unit length of the thread. Also determine the mass of the scale pan.
18. Calculate the frequency n of the given fork with the help of equations (1) and (2) and take the mean.

Experimental Data:

Mass of the scale pan, $w_1 = \dots\dots\dots$ gm

Length of the sample thread, $L = \dots\dots\dots$ cm

Mass of the thread, $M = \dots\dots\dots$ gm

Mass per unit length of the thread, $m = (M/L) = \dots\dots\dots$ gm/cm

Table 1: Data for determination of the frequency of a tuning fork for longitudinal position.

∴	No. of Obs.			
	1	2	3	
				Total no. of loops between the fixed ends
				Load on the scalepan w (gm)
				Mass of the scale pan, w_i (gm)
				Tension, $t = (w + w_i)g$ (dynes)
				Distance between the pins, G (cm)
				No. of loops between the pins (N)
				Length of a segment, $l = G/N$ (cm)
				$\frac{t}{l^2} = 4mn^2 = \text{constant}$
				Frequency of the string, $n = n' = \frac{1}{\lambda} \sqrt{\frac{t}{m}} = \frac{1}{2l} \sqrt{\frac{t}{m}}$
				Frequency of the fork, $n = 2n'$ (vibration per sec)
				Mean frequency (vibration per second)

Table 2: Data for determination of the frequency of a tuning fork for transverse position.

No. of Obs.	Total no. of loops between the fixed ends	Load on the scale pan w (gm)	Mass of the scale pan, w_t	Tension, $t = (w + w_t)g$ (dynes)	Distance between the pins, G (cm)	No. of loops between the pins (N)	Length of a segment, $l = G/N$ (cm)	$\frac{t}{l^2} = \text{constant}$	Frequency of the string, $n = n' = \frac{1}{\lambda} \left[\frac{t}{m} \right] \frac{1}{2l}$	Frequency of the fork, $n = n'$ (vibration per sec)	Mean frequency (vibration per second)
1											
2											
3											
...											

Calculations:

Frequency of a tuning fork for longitudinal position,

(i) For first observation,

$$n = 2n' = \frac{1}{l} \sqrt{\frac{t}{m}} =$$

(ii) For second observation,

$$n = 2n' = \frac{1}{l} \sqrt{\frac{t}{m}} =$$

(iii) For third observation,

$$n = 2n' = \frac{1}{l} \sqrt{\frac{t}{m}} =$$

Mean frequency of a tuning fork for longitudinal position, $n =$

Frequency of a tuning fork for transverse position,

(i) For first observation,

$$n = n' = \frac{1}{l} \sqrt{\frac{t}{m}} =$$

(ii) For second observation,

$$n = n' = \frac{1}{l} \sqrt{\frac{t}{m}} =$$

(iii) For third observation,

$$n = n' = \frac{1}{l} \sqrt{\frac{t}{m}} =$$

Mean frequency of a tuning fork for transverse position, $n =$

So, mean frequency of the tuning fork,

$$= \frac{\text{mean frequency for longitudinal position} + \text{mean frequency for transverse position}}{2}$$

Standard value =

$$\text{Percentage of error} = \frac{\text{standard value} - \text{experimental value}}{\text{standard value}} =$$

Results:

Frequency of tuning fork for

- i. Longitudinal position =vibration per sec
- ii. Transverse position =vibration per sec
- iii. Mean frequency =vibration per sec

Percentage of error =

Precautions:

1.
2.
3.

Discussion:

1.
2.
3.